

Embed a thin perforated stainless steel mesh sleeve (pore size ≥ 4 mm) within the catheter floor lumen, secured to the catheter wall,

Executive Technology Brief

Domain: fluidics_hydraulic

Technology Maturity: ENGINEERING_DEFINITION

Transfer Posture: Sponsored Validation

Problem: Documented problem 'lumen occlusion' concerning multi-lumen drainage catheter. User-submitted engineering problem (session input): "Why do multi-lumen drainage catheters lose drain conductance, and can a wider floor lumen restore flow? (plausibility bounds)". [fda_maude] 100 records retrieved live (1997-2032). Illustrative records: 'MAUDE report 100000: Injury [VITAL SIGNS]'; 'MAUDE report 10000032: Malfunction [PUMP MMT-1780KPK 670G PATHWAY BLACK MG]'. LIMITS STAMPED INLINE: voluntary/regulatory reports; counts are NOT incidence rates; causality UNVERIFIED (Art. XXI.5). [europepmc] 10 records retrieved live (2026-2026). Illustrative records: '<i>E. coli</i> bacterium near corrugated surfaces: near-surface swimming, escape, and hydrodynamic trapping.'; 'Computational Fluid Dynamics Reveals Mass Transfer Limitations in a Pilot-Scale Microbial Electrolysis Cell.'. LIMITS STAMPED INLINE: literature context establishes the problem class, not event rates. Problem-existence basis: live records retrieved.

Mechanism: Perforated metal liner (mesh) used in chemical reactor tubes to prevent fouling while allowing fluid flow through the perforations, maintaining conduit integrity under pressure.

Architecture: PROPOSED architecture (no physical system exists): flow path / lumen architecture; pressure regulation element; sensing / feedback element (if closed-loop); termination interfaces (patient / reservoir) — realizing: Embed a thin perforated stainless steel mesh sleeve (pore size ≥ 4 mm) within the catheter floor lumen, secured to the catheter wall,

Governing Model: Fluid mechanics / hydraulics governing relations proposed for this invention; SYMBOLIC ONLY until inputs are sourced

Key Equations:

FLUID-001: Hagen-Poiseuille (laminar pipe flow). Expression: $Q = (\pi * r^4 * dP) / (8 * \mu * L)$.

Applicability to this invention: APPLICABLE; holds under: Newtonian fluid, laminar regime, rigid circular lumen, fully developed flow. Assumptions: CSF approximated as Newtonian (valid at shunt flow rates); no wall compliance effects. Engaged variables: r , dP , L .

FLUID-002: Reynolds number. Expression: $Re = (\rho * v * D) / \mu$. Applicability to this invention:

APPLICABLE; holds under: circular section; single-phase flow. Assumptions: transition criterion $Re \sim 2300$ is a MODEL_DERIVED engineering rule of thumb. Engaged variables: D .

FLUID-003: Orifice flow (sharp-edged restriction). Expression: $Q = C_d * A * \sqrt{2 * dP / \rho}$.

Applicability to this invention: APPLICABLE; holds under: turbulent orifice flow; C_d treated as MODEL_DERIVED until measured. Assumptions: $C_d \sim 0.6-0.65$ is an engineering prior, not a measurement. Engaged variables: dP .

Kill Condition: killer experiment fails: falsification_test_from_candidate (decision scores: EIG 0.4557178723356231, EIG-per-cost 0.4557178723356231; the falsification test carries the pre-registered pass/fail before any transfer)

Current State: ENGINEERING_DEFINITION. No physical prototype. No clinical validation.

Disclosure: This dossier distinguishes established evidence from proposals and unresolved questions.